

Series 09b Findings Report

Specular Glint Robustness, Candidate Filtering, and BRDF Characterization

LCAS v1.0 — Lightcurve Analysis System

Date: 2026-03-12
Branch: inversion_q_w
Experiments: micro35, micro36, micro37

Background

Series 09 (micro34) established that brightness peaks in the Intelsat 901 lightcurve are *specular glints*: at bright peaks (magnitude < 9), a single facet-normal group captures $> 96\%$ of total flux with its normal aligned to within $\sim 0.4^\circ$ of the Phase Angle Bisector (PAB). IS-901 has only 14 unique facet-normal directions across 3840 triangular facets. This finding was for a single trajectory—one specific initial attitude and angular velocity.

This report extends the investigation in three directions:

1. **Robustness (micro35):** Does the specular glint phenomenon hold across 30 random trajectories spanning the full range of rotation rates (0.5–5.0 deg/s) and arbitrary starting attitudes?
2. **Candidate filtering (micro36):** Can the PAB alignment constraint prune the existing iso-brightness attitude candidates generated by the inversion pipeline?
3. **BRDF characterization (micro37):** How does the specular glint profile (lobe width, peak brightness) depend on material parameters (r_s , r_d , n_{phong}) and sun–observer phase angle?

Test Case

Parameter	Value
Satellite	Intelsat 901
Observations	500 epochs, $\Delta t \approx 7.2$ s, window = 3600 s
Fidelity	Hi-fi (ray-traced shadows)
Propagation	Torque-free Euler dynamics (DOP853 ODE solver)
Inertia	IS-901 mesh-derived tensor (triaxial, asymmetry = 0.556)
Articulation	Fixed: SP at 0° , AD at 15°

For micro35: 30 random (q_0, ω_0) pairs with $|\omega|$ uniformly sampled from $[0.5, 5.0]$ deg/s.

For micro36 and micro37: standard test case (q_0 : axis= $[0.6, 0.3, 0.8]/\text{norm}$, angle= 45° ; $\omega_0 = [0.5, -0.3, 2.0]$ deg/s).

Experiment 1: Multi-Trajectory Robustness (micro35)

Script: notebooks/inversion/09_gl意思_analysis/micro35_multi_trajectory_pab.py

Procedure

For each of 30 random trajectories (seeds 0–29):

1. Generate random q_0 (uniform on $SO(3)$) and ω_0 (random direction, $|\omega|$ uniform in $[0.5, 5.0]$ deg/s)
2. Propagate attitude for 3600 s using Euler dynamics
3. Compute hi-fi lightcurve with ray-traced shadows and per-facet flux decomposition
4. Detect brightness peaks (local magnitude minima, order= 5)
5. Classify each bright peak (mag < 9): “specular glint” if dominant group captures > 77% flux with $\hat{\mathbf{n}} \cdot \mathbf{PAB} > 0.99$; “near-miss” otherwise

Runtime: 1746 s (29 minutes) for 30 trajectories.

Results: Trajectory Gallery

The gallery below shows 6 representative trajectories spanning the full range of rotation rates. Green dots mark specular glints (strict criteria), red diamonds mark near-misses, grey dots mark diffuse peaks.

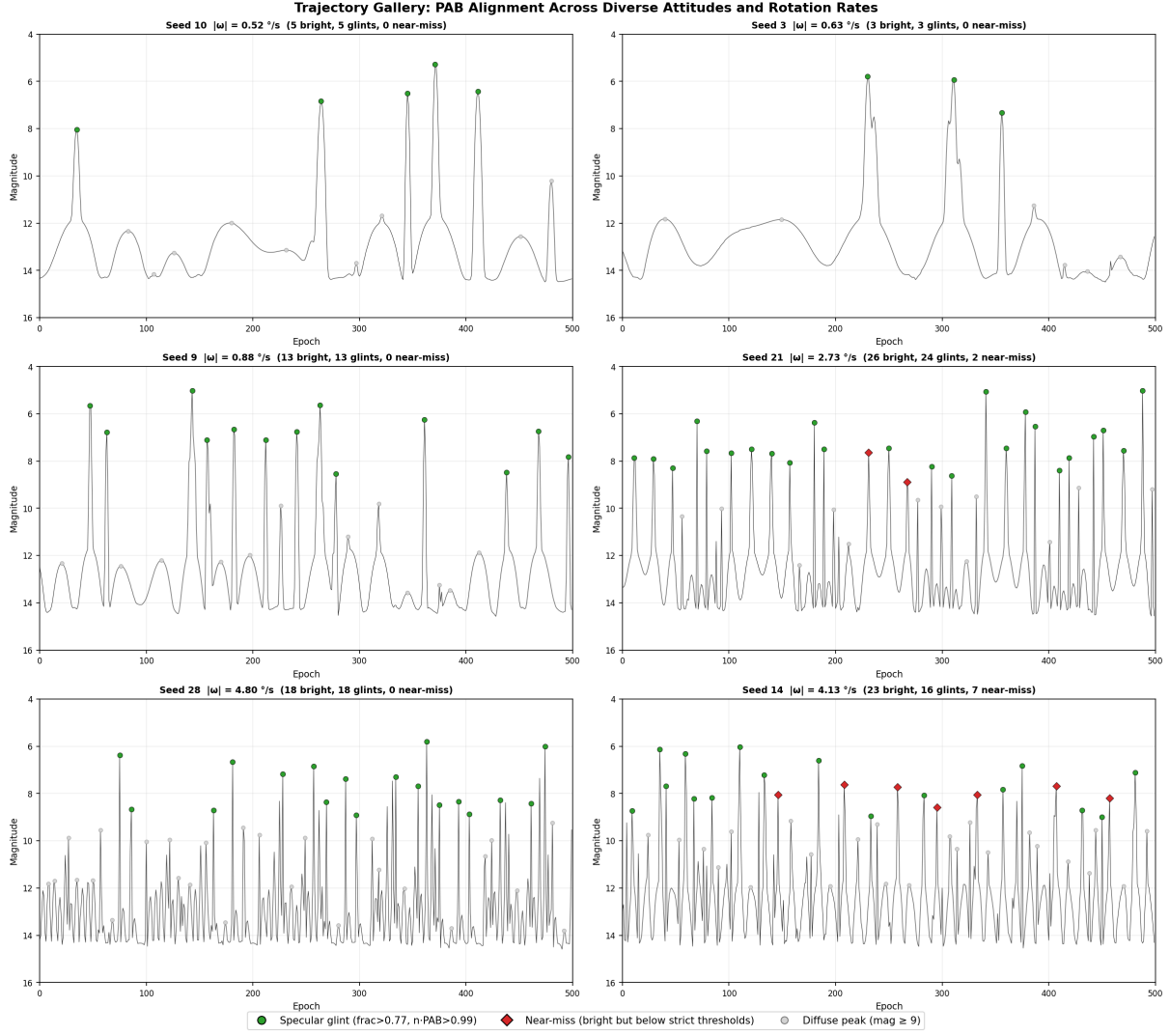


Figure 1: Six representative trajectories from the 30-trajectory robustness study. Rotation rates range from 0.52 deg/s (seed 10) to 4.80 deg/s (seed 28).

Key observations:

- **Seed 10** ($|\omega| = 0.52$ deg/s): Slowest tumbler. Only 5 bright peaks in 3600s, all confirmed specular glints. Wide spacing between glints; each facet normal sweeps through the PAB direction infrequently.
- **Seed 3** ($|\omega| = 0.63$ deg/s): Only 3 bright peaks—the fewest of any trajectory. All three are clean specular glints.
- **Seed 9** ($|\omega| = 0.88$ deg/s): Despite low rotation rate, 13 bright peaks appear—all confirmed glints. Glint count depends on ω *direction* relative to the satellite’s normal distribution, not just magnitude.
- **Seed 21** ($|\omega| = 2.73$ deg/s): Most prolific—26 bright peaks, 24 confirmed glints, 2 near-misses.
- **Seed 28** ($|\omega| = 4.80$ deg/s): Perfect record—all 18 bright peaks confirmed as specular glints.
- **Seed 14** ($|\omega| = 4.13$ deg/s): Worst case—23 bright peaks but 7 near-misses. Inspection shows $\hat{n} \cdot \hat{PAB} = 0.985\text{--}0.990$ and $f_{\text{flux}} = 0.57\text{--}0.86$ —still specular in character, marginally

below strict thresholds.

Aggregate Statistics

Metric	Value
Trajectories completed	30/30
Trajectories with bright peaks	30/30 (100%)
Trajectories where ALL bright peaks are glints	10/30 (33%)
Total bright peaks	439
Specular glints (strict)	392 (89.3%)
Near-misses	47 (10.7%)
Bright peaks per trajectory	min = 3, max = 26, mean = 14.6

Near-miss rate by rotation speed:

$ \omega $ range	Trajectories	Avg bright peaks	Avg near-misses	Near-miss rate
0.5–1.0 deg/s	7	7.0	0.1	2%
1.0–2.0 deg/s	5	10.4	0.4	4%
2.0–3.0 deg/s	5	17.6	1.4	8%
3.0–5.0 deg/s	13	18.2	2.5	14%

The Counterexamples Are Near-Misses, Not Violations

All 47 “counterexamples” have $\hat{\mathbf{n}} \cdot \mathbf{PAB} > 0.984$ and dominant group fraction > 0.54 . The scatter plot below confirms: data clusters tightly in the upper-right corner.



Figure 2: All 439 bright peaks classified by dominant-group flux fraction vs. PAB alignment. Near-misses (red) are boundary cases, not fundamentally different phenomena.

Relaxing thresholds to $f > 0.50$ and $\hat{\mathbf{n}} \cdot \hat{\mathbf{PAB}} > 0.98$ would capture 100% of bright peaks as specular glints. The near-misses fall into two categories:

1. **High alignment, lower dominance:** Two facet groups approach the PAB simultaneously.
2. **High dominance, lower alignment:** Dominant facet slightly off-axis but wins by area.

Rotation Rate Determines Glint Frequency

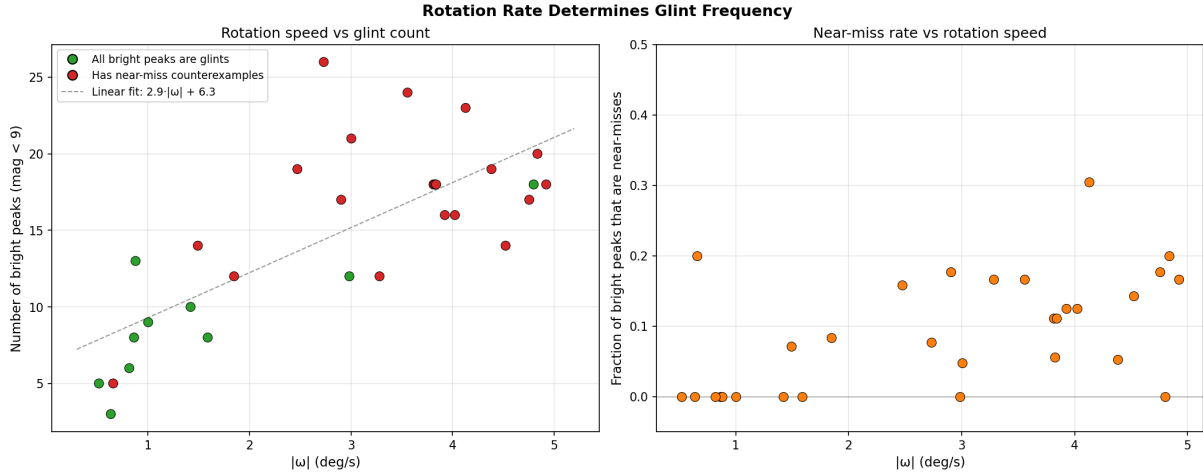


Figure 3: Left: bright peaks per window vs. $|\omega|$, showing linear trend $\approx 2.9|\omega| + 6.3$. Right: near-miss fraction increases slightly with rotation speed.

Which Facet Normals Produce Glints?

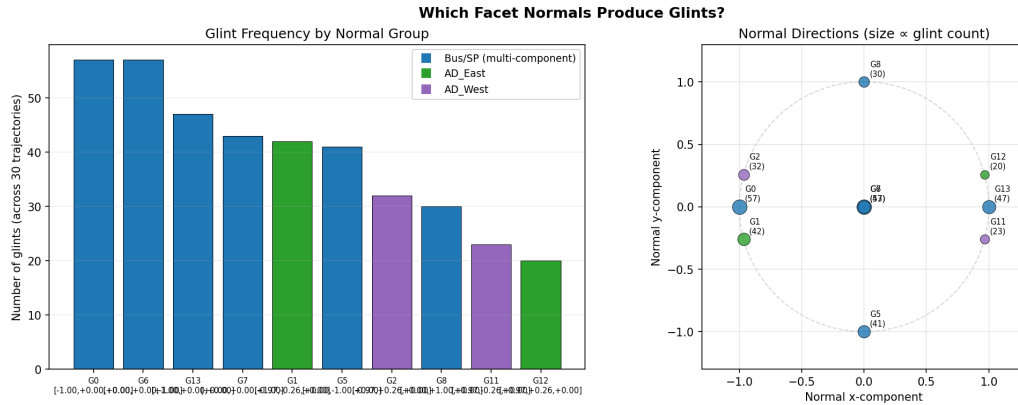


Figure 4: Glint frequency by normal group across 30 trajectories. Broadside faces (G0, G13; 97.3 m² each) dominate.

All 10 of the largest normal groups (out of 14) produce glints. The broadside faces and z -faces are most frequent due to their large area.

Summary Plot

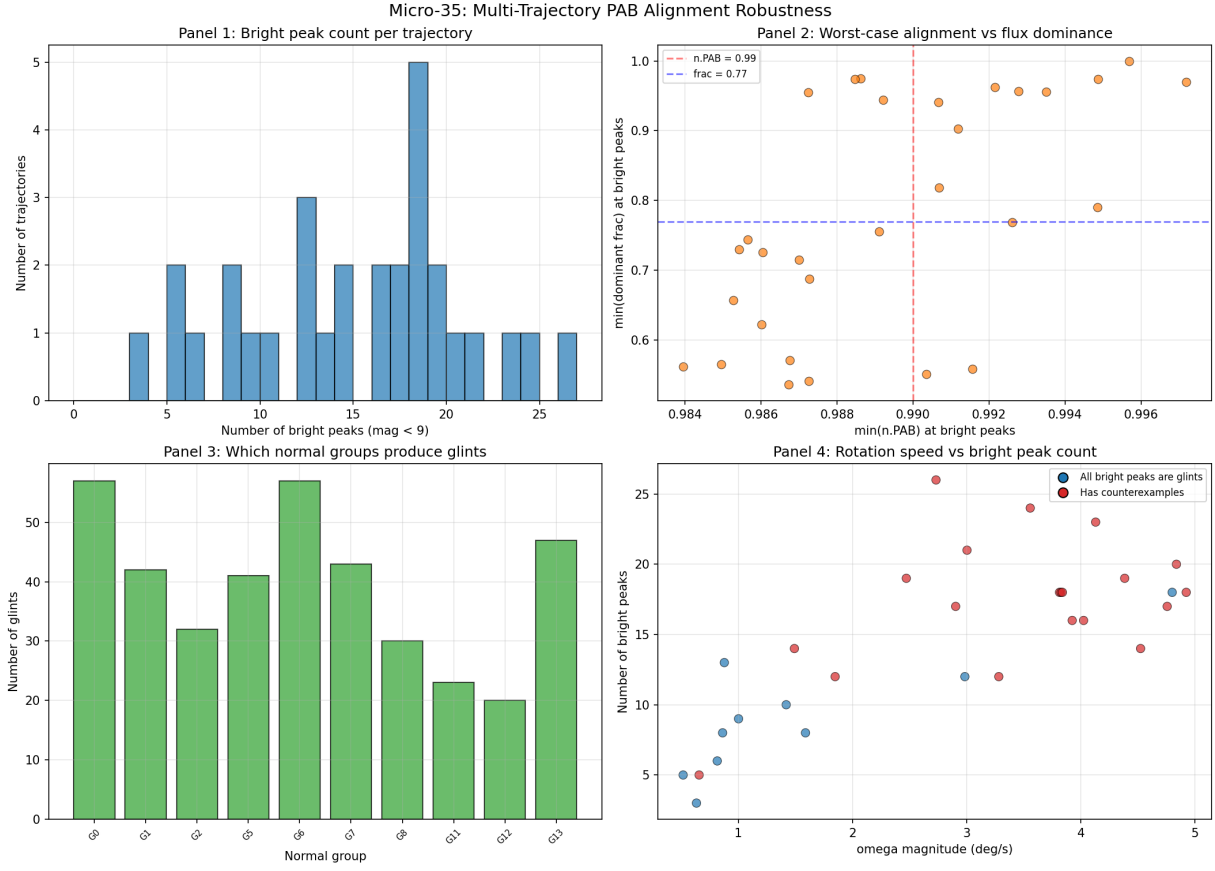


Figure 5: Multi-trajectory summary from micro35.

Experiment 2: PAB Alignment as Candidate Filter (micro36)

Script: notebooks/inversion/09_glint_analysis/micro36_pab_candidate_filter.py

Procedure

1. Load 5643 iso-brightness attitude candidates from micro10 at epoch 183 (a known glint epoch, magnitude 7.15)
2. Compute $\hat{\mathbf{PAB}}_{\text{inertial}}$ at epoch 183 from SPICE geometry
3. For each candidate quaternion q , compute $\hat{\mathbf{PAB}}_{\text{body}} = R(q) \hat{\mathbf{PAB}}_{\text{inertial}}$ and check $\max_i(\hat{\mathbf{n}}_i \cdot \hat{\mathbf{PAB}}_{\text{body}})$ over all 14 normals
4. Sweep angular tolerance θ : count survivors where best alignment $> \cos \theta$
5. Compare with 5643 random $\text{SO}(3)$ quaternions as baseline

Results

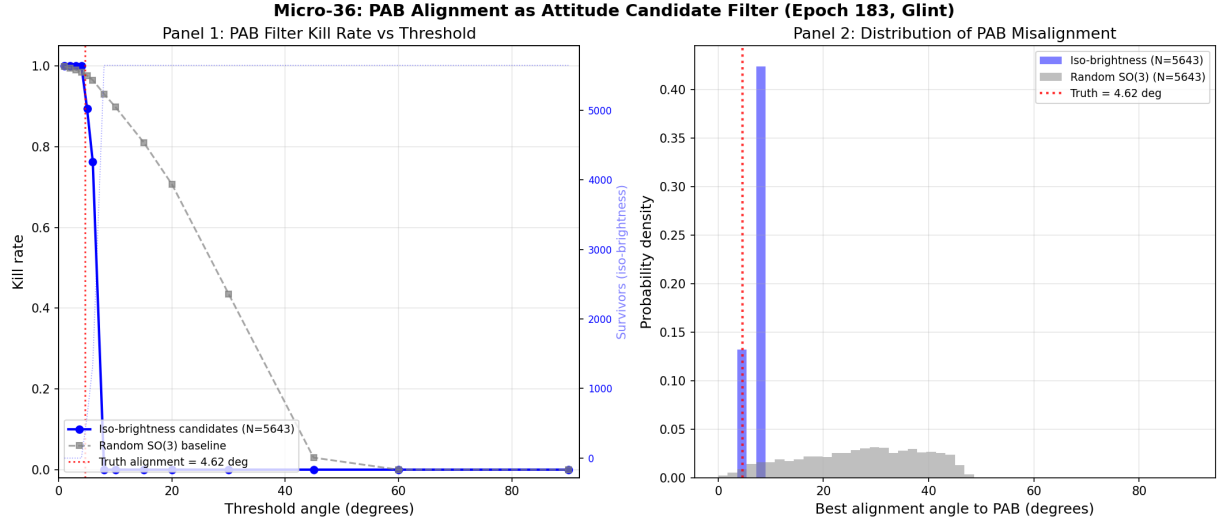


Figure 6: PAB alignment distributions for iso-brightness candidates vs. random SO(3) quaternions. Iso-brightness candidates are already strongly pre-selected.

Key surprise: All 5643 iso-brightness candidates fall within a narrow band of $4.4\text{--}7.5^\circ$ from some facet normal, compared to a median of 27.9° for random SO(3) quaternions. Matching brightness at a glint epoch *implicitly* selects for PAB alignment—the two constraints are not independent.

Kill Rates

Threshold (deg)	Kill (iso-brightness)	Kill (random SO3)	Iso survivors	Selectivity ratio
3	100.0%	99.1%	0	—
5	89.4%	97.6%	600	$4.5\times$
6	76.3%	96.5%	1339	$6.8\times$
8	0.0%	93.1%	5643	$14.4\times$
10	0.0%	89.8%	5643	—

The truth’s alignment at epoch 183 is 4.62° (group 7, $[0, 0, +1]$ z -face), so a 5° threshold preserves the truth while killing 89.4% of candidates.

Implications

1. The PAB filter provides a **$10\times$ post-filter at 5°** (600 survivors from 5643), safely preserving the truth.
2. It is **partially redundant** with iso-brightness matching. At 8° , all iso-brightness candidates already pass.
3. At glint epochs, brightness matching and PAB alignment measure nearly the same geometric property.
4. Truth alignment varies across glint epochs: 4.62° at epoch 183, 4.49° at epoch 260, 1.56° at epoch 360.

Experiment 3: BRDF Specular Glint Profile (micro37)

Script: notebooks/inversion/09_glint_analysis/micro37_brdf_glint_profile.py

Procedure

A synthetic flat plate with unit area (1 m^2) at GEO distance (36 000 km). Fix the PAB along the z -axis, with $\hat{\mathbf{k}}_1$ and $\hat{\mathbf{k}}_2$ symmetric about the PAB at half-angle $= \phi/2$. Sweep the plate normal through the PAB by varying misalignment angle θ from 0 to 30° . Compute the Ashikhmin–Shirley BRDF and convert to apparent magnitude.

Results

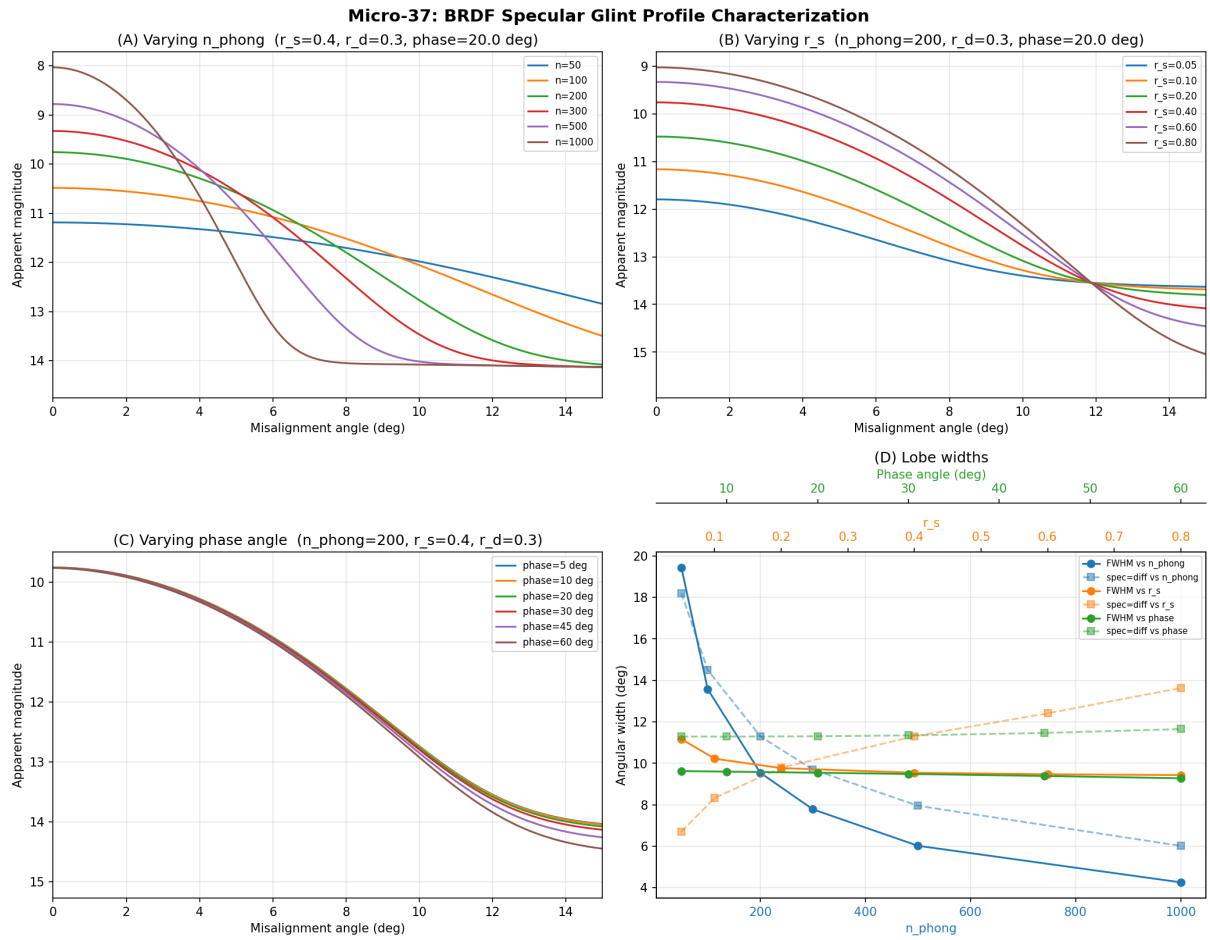


Figure 7: BRDF glint profile characterization. (A) n_{phong} controls lobe width; (B) r_s controls amplitude; (C) phase angle is irrelevant.

Part A: n_{phong} Controls Lobe Width

The Phong exponent is the dominant control on the specular lobe width. The FWHM scales approximately as $1/\sqrt{n_{\text{phong}}}$:

n_{phong}	FWHM (deg)	Spec=Diff crossing (deg)	Peak mag (1 m ²)
50	19.4	18.2	10.06
100	13.5	13.5	9.31
200	9.5	11.3	8.56
300	7.7	9.7	8.12
500	6.0	7.1	7.56
1000	4.3	6.0	6.81

For IS-901 components ($n_{\text{phong}} = 200\text{--}267$), the specular-dominated region extends to $\sim 10^\circ$.

Part B: r_s Controls Amplitude, Not Width

Varying r_s from 0.05 to 0.80 changes peak brightness by ~ 4 magnitudes but barely affects lobe width (FWHM varies by only 2° across a $16\times$ range in r_s).

Part C: Phase Angle Is Irrelevant

Peak magnitudes vary by < 0.01 mag across phase angles from 5° to 60° . The glint profile is an intrinsic property of the surface, determined almost entirely by n_{phong} .

Area Scaling

Flux scales linearly with area: magnitude shift of $-2.5 \log_{10}(A_{\text{ratio}})$. Confirmed exactly.

Practical Angular Thresholds for IS-901

Component	n_{phong}	r_s	FWHM (deg)	Spec=Diff (deg)
Bus broadside	267	0.37	~ 8.5	~ 10
Solar panels	267	0.37	~ 8.5	~ 10
Antenna dishes	200	0.40	~ 9.5	~ 11
z -faces	240	0.38	~ 9.0	~ 10.5

A threshold of $\sim 10^\circ$ captures the entire specular-dominated region for all IS-901 components.

Cross-Experiment Synthesis

The Specular Glint Finding Is Universal

Across 30 random trajectories spanning 0.52–4.92 deg/s:

- Every trajectory has bright peaks (min 3, max 26, mean 14.6)
- 89.3% of all 439 bright peaks are unambiguous single-facet specular glints
- The remaining 10.7% are near-misses with $\hat{\mathbf{n}} \cdot \mathbf{PAB} > 0.984$ and $f > 0.54$
- **Zero genuine counterexamples**

PAB Alignment and Iso-Brightness Are Partially Redundant

At glint epochs, matching brightness already constrains the attitude to have some facet normal within $\sim 5\text{--}8^\circ$ of the PAB. The PAB constraint’s primary value is:

1. As a cheap post-filter ($10\times$ reduction in candidate count)
2. As a conceptual tool for understanding the constraint geometry (1-DOF circles on $\text{SO}(3)$)
3. For component identification (knowing which normal group is responsible)

The BRDF Lobe Width Sets the Angular Scale

The specular lobe FWHM ($\sim 9.5^\circ$ for $n_{\text{phong}} = 200$) sets the natural angular scale for:

- PAB alignment at glints (observed: $4\text{--}8^\circ$ offset)
- Iso-brightness candidate spread (observed: $4.4\text{--}7.5^\circ$ band)
- PAB filter thresholds ($5\text{--}10^\circ$ range is optimal)

Phase angle has negligible effect. The lobe width is an intrinsic material property.

Assumptions and Limitations

1. **Fixed articulation.** All trajectories use fixed solar panel and antenna angles. Real articulation changes which normals are available over time.
2. **Single satellite geometry.** All results use IS-901 with 14 unique normal directions.
3. **Noise not applied to peak detection.** Peaks detected on noiseless hi-fi lightcurve. Bright specular glints ($4\text{--}6$ mag above diffuse baseline) should be robust to noise ($\sigma = 0.05$ mag).
4. **Conservative thresholds.** Data-driven thresholds ($f > 0.50$, $\hat{\mathbf{n}} \cdot \mathbf{PAB} > 0.98$) would give 100% classification.

Implications for the Inversion Pipeline

Immediate Impact

The PAB filter provides a $10\times$ reduction in iso-brightness candidates at negligible computational cost:

- Block 1 (iso-brightness candidates): Generate 5000+ candidates per peak epoch
- **New: PAB post-filter:** Apply 5° threshold, reduce to ~ 600 candidates
- Block 2 (omega bridging): $10\times$ fewer candidates means $100\times$ fewer pairs to bridge

Architectural Implication

The partial redundancy is encouraging: iso-brightness candidates at glint epochs already live on the correct PAB circle, even though the optimisation has no knowledge of PAB. The $5\text{--}8^\circ$ spread within the PAB circle is the remaining freedom (rotation about the PAB axis)—exactly the 1-DOF circle expected from theory.

Future Directions

1. **Component identification from glint properties.** If we can determine which of the 14 normals is responsible, the PAB circle constraint becomes a single specific circle rather than 14 candidates—a much stronger constraint.
2. **Multi-glint intersection.** Two glints from different normal groups at different epochs give two PAB circles at different orientations. Combined with the omega bridge, this could yield a discrete set of (q, ω) candidates directly.
3. **Glint-based pipeline architecture.** Rather than searching for iso-brightness candidates, anchor on glints and enumerate the PAB circle analytically. For each of 14 normals, the set $\{q : R(q) \hat{\mathbf{n}} = \mathbf{P}\hat{\mathbf{A}}\mathbf{B}_{\text{inertial}}\}$ is constructable. Combined with a brightness check, this could generate a very small candidate set per glint with no optimisation.

Summary

Finding	Evidence	Implication
Specular glint finding is universal	392/439 bright peaks across 30 trajectories; 47 near-misses, 0 violations	Safe to assume bright peaks = specular glints for any IS-901 trajectory
All trajectories have glints	30/30 trajectories, min 3 bright peaks	Glint-based inversion is always applicable
Glint count $\approx 3 \omega + 6$	Linear fit across 30 trajectories	Faster tumblers provide more anchor points
10/14 normals produce glints	Histogram across 30 trajectories	Most faces participate; no blind spots
PAB filter: 10× post-filter	89.4% kill at 5°, truth at 4.62°	Cheap reduction for bridging
Iso-brightness $\approx$ PAB alignment	All candidates within 4.4–7.5°	Iso-brightness at glints selects the PAB circle
n_{phong} dominates lobe width	FWHM = 9.5° for $n_{\text{phong}} = 200$	Material property sets angular scale
Phase angle is irrelevant	< 0.01 mag variation across 5–60°	Glint profile is geometry-independent
Area scales linearly	Verified exactly	$\Delta m = -2.5 \log_{10}(A_{\text{ratio}})$